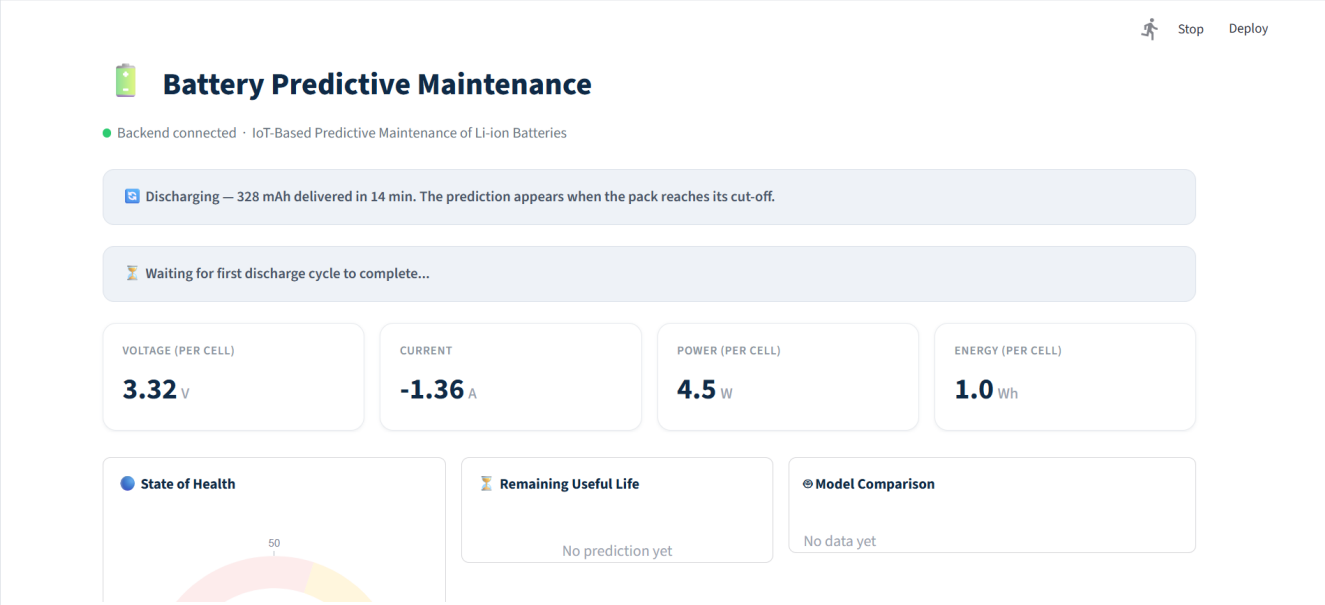
What the dashboard is showing

Your first live discharge of the 3-cell pack – 7 October 2026, about 09:05–09:20 local time

In one sentence
The whole system is working: the pack is discharging through the resistors at about 1.36 A, the PZEM-017 is measuring it, the ESP32 is sending a reading every 5 seconds, the backend is counting the charge delivered, and the dashboard is showing it all live. No SOH/RUL prediction yet – that only appears when the pack reaches its cut-off and the discharge is complete.

1. Top of the dashboard: status, banners and live readings

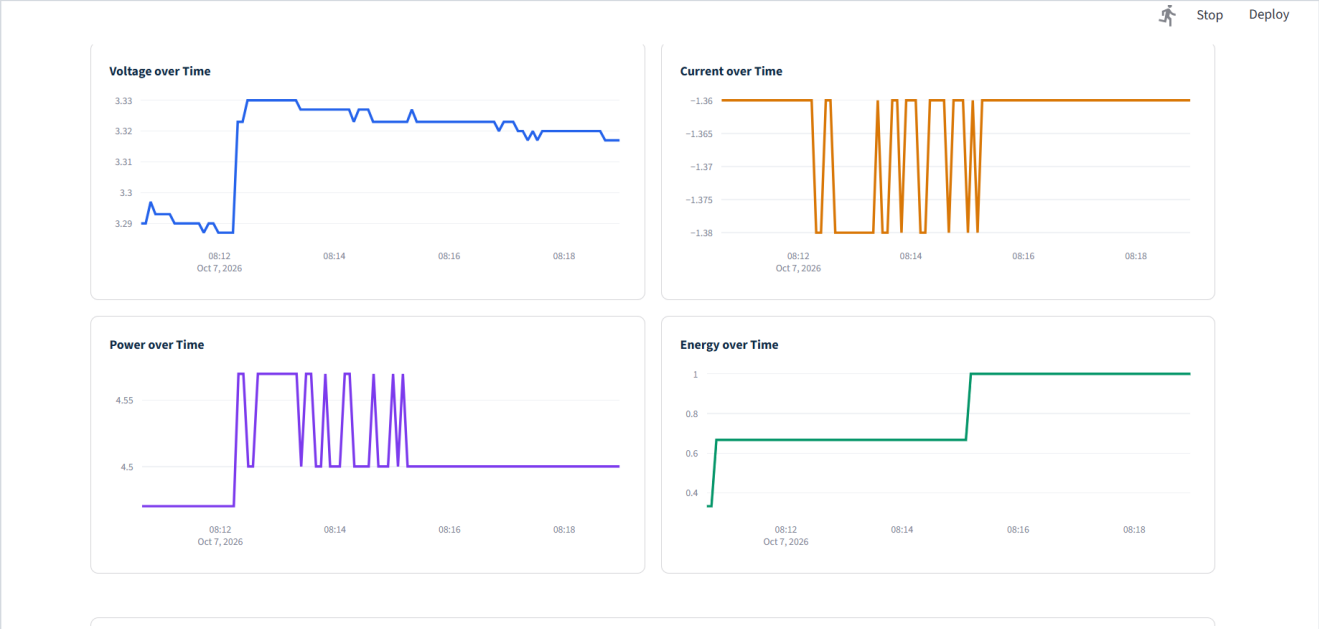


Screenshot 1 – status line, banners, live metric cards and the (still empty) prediction panels.

What you see	What it means
● Backend connected	The dashboard can reach the backend on the laptop. If this turned red, the backend window was closed.
Blue banner: Discharging — 328 mAh delivered in 14 min	The backend has detected a discharge (current below −0.5 A) and is adding up the charge leaving the pack: 1.36 A for 14 minutes ≈ 330 mAh, so the count is correct. This number becomes the **measured SOH** at the end (mAh ÷ 2000 × 100).
Grey banner: Waiting for first discharge cycle to complete…	No discharge has finished yet in this database, so there is no prediction to show. Normal.
VOLTAGE (PER CELL) 3.32 V	The pack voltage divided by 3 (the pack is ≈ 9.96 V). Per-cell values are what the models were trained on.
CURRENT −1.36 A	The discharge current. **Negative = discharging** (current leaving the pack). Check: 9.96 V ÷ 6.6 Ω ≈ 1.5 A, slightly less because of wire resistance and the resistors heating up.
POWER (PER CELL) 4.5 W	Voltage × current per cell: 3.32 V × 1.36 A ≈ 4.5 W. The whole pack delivers ≈ 13.5 W, all turned into heat in the resistors.
ENERGY (PER CELL) 1.0 Wh	The PZEM-017's own energy counter (it counts whole Wh, so 3 Wh for the pack ÷ 3 = 1.0). It is only for display: the backend calculates energy far more precisely from voltage × current × time, and that value goes to the models.

What you see	What it means
State of Health / Remaining Useful Life / Model Comparison: empty	**Deliberately empty until the pack is empty.** The models only understand complete discharges, so the prediction is made once, at the end.

2. The charts



Screenshot 2 – voltage, current, power and energy over the last 100 readings.

Chart	What it shows	Why it looks like that
Voltage over Time	Per-cell voltage between 3.29 and 3.33 V – almost flat.	A Li-ion cell holds a fairly steady voltage for most of a discharge, then drops quickly near the end. The small rise around 08:12 (09:12 local) happened as the current fell slightly: less current means a smaller voltage drop inside the cells, so the terminal voltage rises a little.
Current over Time	Jumping between −1.36 and −1.38 A.	**Not a real fault.** The PZEM-017 measures current in 0.01 A steps, so a current sitting near −1.37 A flips between two neighbouring values. The chart's scale covers only 0.02 A, which makes the flicker look large. The slow drift from −1.38 to −1.36 A is real: the cement resistors heat up and their resistance rises slightly.
Power over Time	Stepping between 4.5 and 4.6 W.	Same cause: power follows the current, and the PZEM reports it in small fixed steps.
Energy over Time	A staircase: 0.33 → 0.67 → 1.0 Wh.	The PZEM's energy counter only advances in whole watt-hours for the pack (1 Wh ÷ 3 cells = 0.33 Wh per step). This is exactly why the backend calculates energy itself.

Times on the charts were 1 hour behind – now fixed
The backend stores time in UTC; your laptop is on Nigeria time (UTC+1). That is why the charts said 08:12–08:18 while 'Last updated' said 09:19. The dashboard now converts to local time: click **Rerun** (top-right of the dashboard) or refresh the browser tab to see it. The measurements themselves were never affected.

3. Raw Readings table



Screenshot 3 – the most recent readings received from the ESP32.

Each row is one reading from the ESP32, about every 5 seconds. Across these rows the voltage is **3.317 V per cell** and the current **−1.36 A** – steady, as expected in the middle of a discharge. The **energy** column stays at 1 for the same 1 Wh-counter reason as the chart. A gap of more than about 10 seconds between rows would mean readings were missed – here there are none.

4. What happens next in this discharge

1. The voltage stays near 3.3 V per cell for a while, then **falls more quickly** as the cells run out.
2. When any cell reaches about 2.5–2.8 V, the **BMS disconnects the load**. The current drops to about 0 A.
3. The backend sees the current rise above −0.5 A, **closes the cycle**, checks that the voltage went low enough (≤ 3.2 V per cell) and runs the models.
4. The dashboard then shows the **SOH gauge, RUL, alert banner and model comparison**, plus a new row: **Charge delivered · Measured SOH · Predicted SOH · Prediction error**.
5. **Switch the switch OFF** once the BMS has cut off, and let the resistors cool before touching anything.

> **Read this run's result with care – it is a rehearsal**
> The pack did **not** start from a full charge, and the first minutes happened before the backend was started. So the **charge delivered and measured SOH will come out lower than the true health** of the cells, and the prediction may not match. This run proves the system works end to end; it is not your result for Table 4.8.

5. The official test (your real result)

1. Take the cells out and charge **each one** on its TP4056 until the light turns blue (full).
2. Rest them 30 minutes; check each reads **4.15–4.20 V** and all are within 0.05 V of each other.
3. Put them back in the holders. Double-click **start_system.bat** and wait for “models loaded”.
4. **Only then** switch ON. Leave it until the BMS cuts off (about 75–90 minutes for good cells).
5. Record the dashboard's **charge delivered, measured SOH, predicted SOH, prediction error, RUL and alert**, and take a screenshot for the report. Repeat with the worn cells.

> **Safety while it runs**
> The resistors are above 100 °C – do not touch them • feel the cells now and then: warm is fine, hot means switch OFF at once • keep the laptop awake and the backend window open • never leave the rig unattended.